Dimensional Model Architecture Report: Zillow Real Estate Data

1. Modeling Rationale: Why This Architecture?

The raw Zillow datasets track metrics at entirely different geographic granularities (City, Metro/MSA, and Zip Code). Joining these raw tables directly would create massive data duplication or drop records because their native region_id identifiers do not align.

The Solution: A Star Schema

This architecture utilizes a Star Schema to resolve the granularity mismatch. We extract the shared geographic concept—the **Metro area**—into a central, conformed dimension table (dim_geography). The granular fact tables are aggregated up to this shared Metro level and linked via a surrogate key (geo_key).

Key Design Decisions:

- **Surrogate Keys for Facts:** Each fact table now includes its own primary key (house_value_id, monthly_payment_id, rental_income_id). This is excellent for data lineage, auditing, and managing row-level updates without relying solely on composite keys.
- **Enriched Dimension:** The dim_geography table standardizes location data and now includes county_name. The ETL explicitly uses a COALESCE function during upserts to ensure that if one source table lacks county data, it doesn't overwrite existing county data with a NULL value.
- **Idempotent Upserts:** All tables utilize ON CONFLICT constraints paired with updated_at = current_timestamp to ensure the ETL can run repeatedly without duplicating data, while accurately tracking when a record was last modified.

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2. Table Schemas

Dimension Table: dim_geography

Acts as the single source of truth for location data, translating disparate region definitions into a unified Metro identity.

Column	Data Type	Key Type	Description

geo_key	serial	**Primary Key**	Surrogate auto-incrementing key used to join to fact tables.
metro	text	Unique	The standardized Metropolitan Statistical Area name (e.g., "Seattle-Tacoma-Bellevue, WA").
state	text	None	The state abbreviation.
county_name	text	None	The county name associated with the metro area.
inserted_at	timestamp	None	Record creation timestamp.
updated_at	timestamp	None	Record modification timestamp.

Fact Table: fact_house_value

Stores the aggregated average home values at the Metro/Date grain.

Column	Data Type	Key Type	Description
house_value_id	serial	**Primary Key**	Unique identifier for the specific fact record.
geo_key	int4	**Foreign Key**	References dim_geography(geo_key).
avg_house_value	float8	Measure	The rolled-up average house value for the metro area.
metric_date	date	Logical Key	The date the metric was recorded. Unique when paired with geo_key.
year	int4	None	The extracted year for partitioning or quick BI filtering.
inserted_at	timestamp	None	Record creation timestamp.
updated_at	timestamp	None	Record modification timestamp.

Fact Table: fact_monthly_payment

Stores the aggregated monthly payments at the Metro/Date grain.

Column	Data Type	Key Type	Description
monthly_payment_id	serial	**Primary Key**	Unique identifier for the specific fact record.
geo_key	int4	**Foreign Key**	References dim_geography(geo_key).
monthly_payment	float8	Measure	The average monthly payment for the metro area.
metric_date	date	Logical Key	The date the metric was recorded. Unique when paired with geo_key.
year	int4	None	The extracted year for partitioning or quick BI filtering.
inserted_at	timestamp	None	Record creation timestamp.
updated_at	timestamp	None	Record modification timestamp.

Fact Table: fact_rental_income

Stores the aggregated rental income at the Metro/Date grain.

Column	Data Type	Key Type	Description
rental_income_id	serial	**Primary Key**	Unique identifier for the specific fact record.
geo_key	int4	**Foreign Key**	References dim_geography(geo_key).
rental_income	float8	Measure	The rolled-up average rental income for the metro area.
metric_date	date	Logical Key	The date the metric was recorded. Unique when paired with geo_key.
year	int4	None	The extracted year for partitioning or quick BI filtering.
inserted_at	timestamp	None	Record creation timestamp.

updated_at	timestamp	None	Record modification timestamp.

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3. Join Strategy & Usage

Because all business metrics have been standardized to the geo_key and metric_date grain, querying across the different real estate domains is highly efficient.

When building views or analytical queries, always start from dim_geography (and potentially a conformed Date dimension if you add one later). Use LEFT JOIN on the fact tables to ensure you do not drop a geographic region just because it happens to be missing a specific month's data in one of the source files.

Sample Analytical Query:

```
SELECT
    g.metro,
    g.state,
    g.county_name,
    fhv.metric_date,
    fhv.avg_house_value,
    fmp.monthly_payment,
    fri.rental_income
FROM dim_geography g
LEFT JOIN fact_house_value fhv
    ON g.geo_key = fhv.geo_key
LEFT JOIN fact_monthly_payment fmp
    ON g.geo_key = fmp.geo_key
    AND fhv.metric_date = fmp.metric_date
LEFT JOIN fact_rental_income fri
    ON g.geo_key = fri.geo_key
    AND fhv.metric_date = fri.metric_date
WHERE g.state = 'WA'
ORDER BY g.metro, fhv.metric_date DESC;
```